# ARYAMAN SRIVASTAVA

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## PROFILE

Focused on cybersecurity, with practical experience building threat-hunting log workflows and cryptographic demonstrations. Combines Python implementation and integration testing with language-model research, Linux troubleshooting and clear technical communication to investigate security problems.

## EDUCATION

**Nanyang Technological University** Singapore | Aug 2026 - Present
MSc Cyber Security; coursework: Cryptography, Application Security, Security Monitoring and Threat Detection.

**Vellore Institute of Technology** Vellore, India | 2021 - 2025
BTech in Computer Science and Engineering; coursework: Operating Systems, DBMS, Computer Networks and Data Privacy.

## EXPERIENCE

**Data Science Intern, Directorate General of Hydrocarbons** Noida, India | Jul - Nov 2025
Ministry of Petroleum and Natural Gas, Government of India

- Fine-tuned language models using LoRA on oil and exploration data, applying parameter-efficient adaptation to domain-specific information.
- Evaluated model outputs through iterative experiments to assess responses on oil and exploration data and guide subsequent fine-tuning.

**Software Development Intern, National e-Governance Division** New Delhi, India | Sep - Nov 2023
Ministry of Electronics and Information Technology, Government of India

- Developed and improved frontend components for a production React Native application supporting government digital public-service workflows.
- Contributed to application debugging and usability improvements; added asynchronous components to the React Native login screen to improve app responsiveness in low-connectivity areas.

## PROJECTS AND LEADERSHIP

**TRACE — Agentic Threat Hunting** NTU | Oct 2026

- Implemented Python schema generation, validation and reuse for unfamiliar log formats, supporting investigations in a team-built, ELK-based agentic threat-hunting system.
- Developed secure source onboarding with duplicate-safe ingestion and audit logs; tested Elasticsearch and MCP integration for threat-hunting investigations.

**The Hidden Colouring — Zero-Knowledge Proofs** NTU | Oct 2026

- Built a Python-backed interactive demonstration of graph three-colourability using SHA-256 commitments and challenge-response verification.
- Implemented public-transcript verification and tamper tests to illustrate hidden-witness privacy and the risks of reusing commitments.

**Healthcare Q&A language-model research** VIT | Dec 2024 - Jun 2025

- Fine-tuned small language models using LoRA for healthcare question answering, designing and evaluating iterative experiments.
- Improved Q&A accuracy from 50.0% to 79.8%, a 29.8 percentage-point gain.

**Linux User Group VIT, Core Committee Member** Mar 2022 - Jun 2025

- Led Linux administration workshops, troubleshot installations and collaborated on accepted open-source pull requests.

## CERTIFICATIONS

AWS Certified Solutions Architect - Associate (Mar 2026)
Machine Learning Specialization, DeepLearning.AI (2023)

## TECHNICAL SKILLS

**Cybersecurity:** Application and network security, cryptography, security monitoring, data privacy.
**Software development:** Python, Kotlin, Java, JavaScript, SQL/DBMS, React Native, REST APIs, Bash, Git.
**Cloud and tools:** AWS, Linux, Docker, GitHub Actions; Microsoft Office (Excel, Word, PowerPoint).
**AI/LLM and analytics:** Language-model evaluation, LoRA fine-tuning, experiment design, Python data analysis.

## SOFT SKILLS

Technical communication, evidence-based reporting, workshop facilitation, teamwork and problem-solving.